

8.4 Shear



Core skills you will master in this section:

- ▶ Shear keeps the coordinate along one direction unchanged while proportionally adjusting the coordinate along another direction.
- ▶ The shear angle controls the direction and strength of the shear, applicable to both horizontal and vertical shear.
- ▶ Consecutive shears along the same axis combine into a single shear.
- ▶ A shear matrix is invertible, and its inverse is still upper triangular or lower triangular.
- ▶ A diagonal square matrix whose main-diagonal elements are all nonzero can be written in either of two forms: "shear @ scaling" or "scaling @ shear".
- ▶ The order of consecutive shear operations along the same axis can be swapped — that is, they satisfy the commutative property of matrix multiplication.
- ▶ In three dimensions, a shear along one axis can depend on either of the other two axes, giving six basic types of shear.

Shear (shear) keeps the coordinates along certain axis directions unchanged, while applying a proportional transformation along other directions. Shear transformations can generally be carried out via matrix multiplication, and are widely used in fields such as computer graphics, linear algebra, and physical simulation. Later in this book, you will encounter shear operations in topics such as Cholesky decomposition and LDL decomposition.

Horizontal Shear

Horizontal shear (horizontal shear) tilts a shape in the horizontal direction, while the x_2 coordinate remains unchanged. The corresponding transformation matrix is

$$\mathbf{K}_{x_1} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$$

where k is the horizontal shear factor, that is, the ratio of horizontal offset.

Clearly, the transformation matrix in the equation above is upper triangular, and its determinant is 1; this means the geometric transformation does not change area.

The coordinates before the transformation correspond to the vector $[x_1, x_2]^T$, and after the shear above, the vector becomes $[y_1, y_2]^T$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + kx_2 \\ x_2 \end{bmatrix}$$

After completing the horizontal shear, the point (x_1, x_2) moves to $(x_1 + k \cdot x_2, x_2)$. That is, the horizontal-axis coordinate x_1 is adjusted to $x_1 + k \cdot x_2$; the vertical-axis coordinate remains unchanged.

⚠ Note that k is not a translation amount, but a multiplier; the equation above tells us that the magnitude of x_2 affects the amount of translation, and the sign of x_2 affects the direction of translation.

The shear factor k controls the direction and strength of the shear. Looking at Figure 1 (a), we can see that when $k > 0$, the shape tilts to the right; while Figure 1 (b) tells us that when $k < 0$, the shape tilts to the left. Clearly, the larger the absolute value of k , the stronger the shear, and the more pronounced the tilt of the shape.

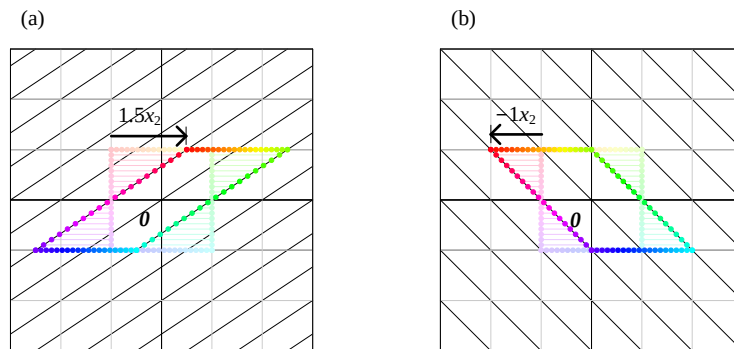


Figure 1. Two Examples of Horizontal Shear

Vertical Shear

Vertical shear (vertical shear) tilts a shape in the vertical direction, while the x_1 coordinate remains unchanged. The corresponding transformation matrix is:

$$K_{x_2} = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

The matrix in the equation above is lower triangular, and its determinant is also 1; this likewise means the geometric transformation does not change area.

After the vertical shear, the vector becomes

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ kx_1 + x_2 \end{bmatrix}$$

As shown in Figure 2, after completing the vertical shear, the point (x_1, x_2) moves to $(x_1, x_2 + k \cdot x_1)$. That is, the horizontal-axis coordinate remains unchanged; the vertical-axis coordinate x_2 is adjusted to $x_2 + k \cdot x_1$.

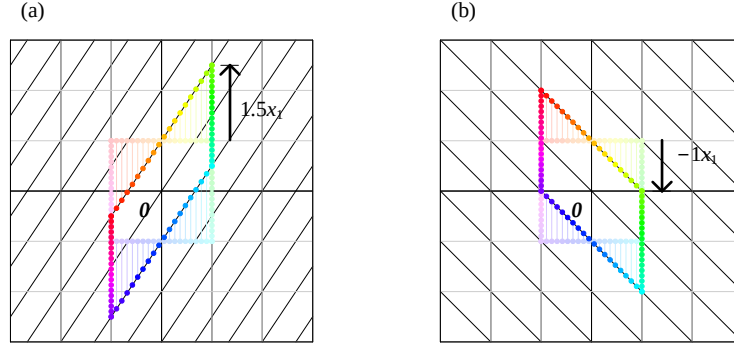


Figure 2. Two Examples of Vertical Shear

Shear Angle

We can also use the shear angle θ to describe planar shear transformations.

The matrix operation corresponding to horizontal shear is

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \cot(\theta) \\ 0 & 1 \end{bmatrix}}_S \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

where $\cot(\theta)$ is the cotangent of the shear angle. The equation above means that the point (x_1, x_2) moves to $(x_1 + \cot(\theta) \cdot x_2, x_2)$. That is, the horizontal-axis coordinate x_1 is adjusted to $x_1 + \cot(\theta) \cdot x_2$; the vertical-axis coordinate remains unchanged.

As shown in Figure 3, in horizontal shear, the shear angle θ controls the strength and direction of the shear. When $0^\circ < \theta < 90^\circ$, the shape tilts to the right. When $0^\circ > \theta > -90^\circ$, the shape tilts to the left.

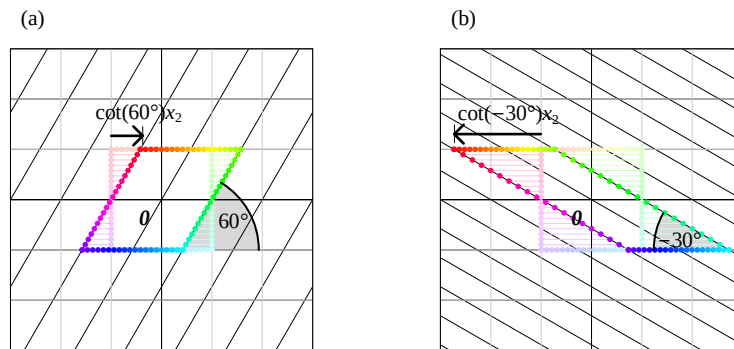


Figure 3. Two Examples of Horizontal Shear, Using the Shear Angle

Described using the shear angle θ , the matrix operation corresponding to vertical shear is

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ \cot(\theta) & 1 \end{bmatrix}}_S \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

After completing the vertical shear, the point (x_1, x_2) moves to $(x_1, x_2 + \cot(\theta) \cdot x_1)$. That is, the horizontal-axis coordinate remains unchanged; the vertical-axis coordinate x_2 is adjusted to $x_2 + \cot(\theta) \cdot x_1$.

As shown in Figure 4, in vertical shear, the shear angle θ likewise controls the strength and direction of the shear. When $0^\circ < \theta < 90^\circ$, the shape tilts upward. When $0^\circ > \theta > -90^\circ$, the shape tilts downward.

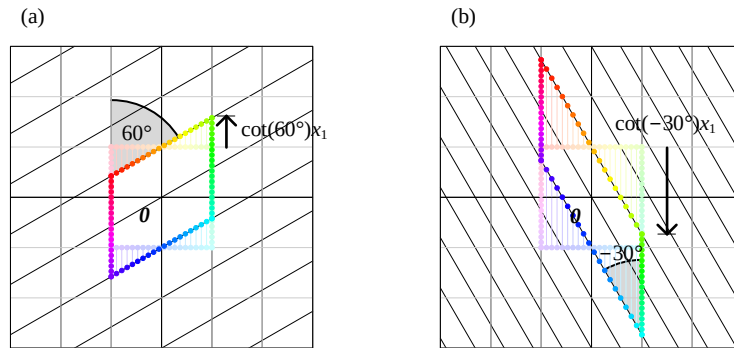


Figure 4. Two Examples of Vertical Shear, Using the Shear Angle

The Inverse of a Shear

The inverse transformation matrix of a shear transformation can be obtained by taking the matrix inverse.

The inverse of the shear matrix along the x_1 axis

$$(K_{x_1})^{-1} = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -k \\ 0 & 1 \end{bmatrix}$$

That is

$$\begin{bmatrix} 1 & -k \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & -k \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Please analyze this example in Figure 5 on your own.

Hopefully you remember, from earlier in this book, that the inverse of an upper triangular matrix is also upper triangular.

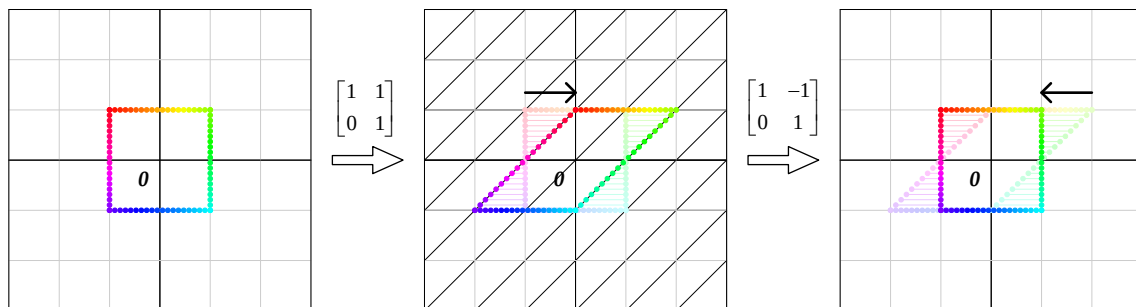


Figure 5. Shear Along the Horizontal Axis, and Its Inverse Operation

The inverse of the shear matrix along the x_2 axis

$$(K_{x_2})^{-1} = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -k & 1 \end{bmatrix}$$

The inverse of a lower triangular matrix is also lower triangular – please review this point as well.

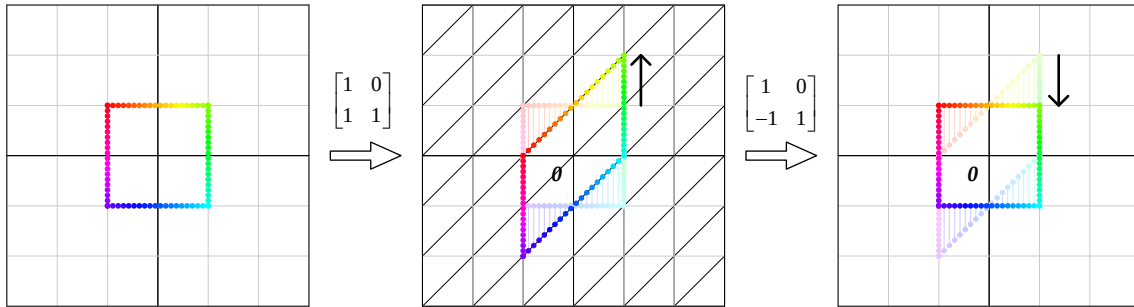


Figure 6. Shear Along the Vertical Axis, and Its Inverse Operation

Consecutive Shears Along the Same Axis

Suppose we apply two consecutive shears along the same axis to a point.

For example, shearing twice along the x_1 axis, the corresponding matrix multiplication is

$$\begin{bmatrix} 1 & k_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & k_2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & k_1 + k_2 \\ 0 & 1 \end{bmatrix}$$

After swapping the order

$$\begin{bmatrix} 1 & k_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & k_1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & k_1 + k_2 \\ 0 & 1 \end{bmatrix}$$

The equation above tells us that swapping the order of same-axis shears does not change the result.

⚠ Note that and are a special case where matrix multiplication is commutative.

Figure 7 shows consecutive shears along the horizontal axis.

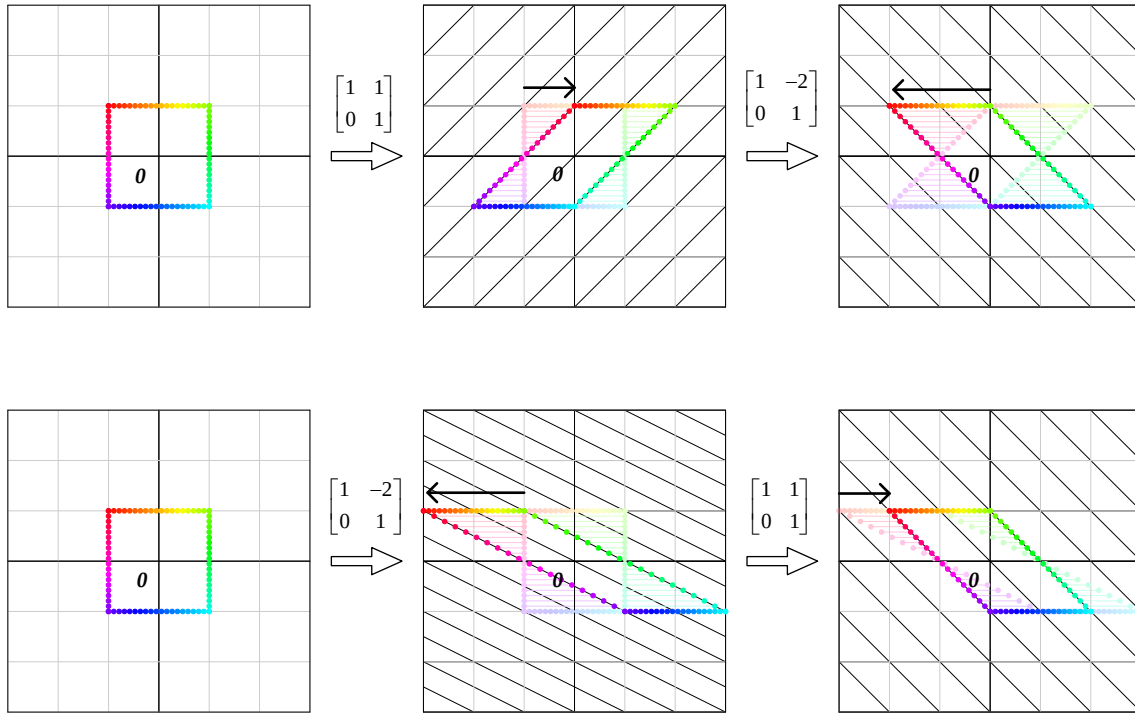


Figure 7. Consecutive Shears Along the Horizontal Axis in the Plane

As another example, shearing twice along the x_2 axis corresponds to

$$\begin{bmatrix} 1 & 0 \\ k_1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ k_2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ k_1 + k_2 & 1 \end{bmatrix}$$

After swapping the order

$$\begin{bmatrix} 1 & 0 \\ k_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ k_1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ k_1 + k_2 & 1 \end{bmatrix}$$

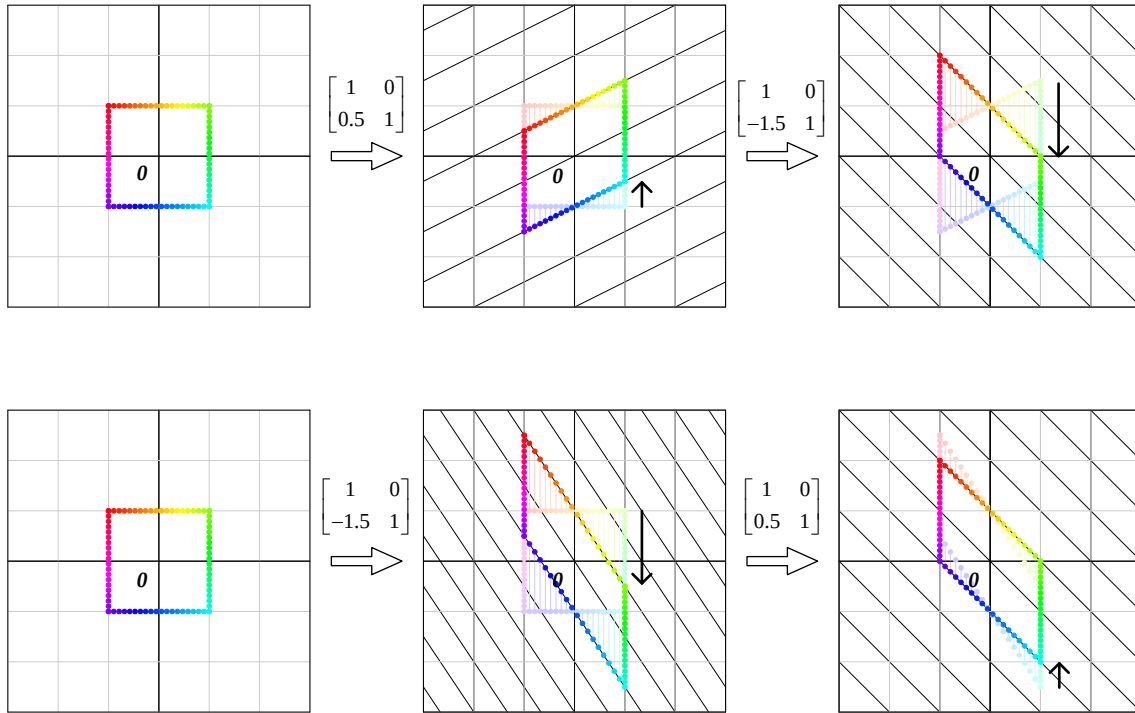


Figure 8. Consecutive Shears Along the Vertical Axis in the Plane

Consecutive Shears Along Different Axes

Figure 9 shows consecutive shears along different axes, giving different results.

Figure 9 (a) First shearing along x_1 , then along x_2

$$\begin{bmatrix} 1 & 0 \\ k_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & k_1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & k_1 \\ k_2 & k_1 k_2 + 1 \end{bmatrix}$$

Figure 9 (b) shearing first along x_2 , then along x_1 instead

$$\begin{bmatrix} 1 & k_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ k_2 & 1 \end{bmatrix} = \begin{bmatrix} k_1 k_2 + 1 & k_1 \\ k_2 & 1 \end{bmatrix}$$

Figure 9 (a) and Figure 9 (b) give completely different results.

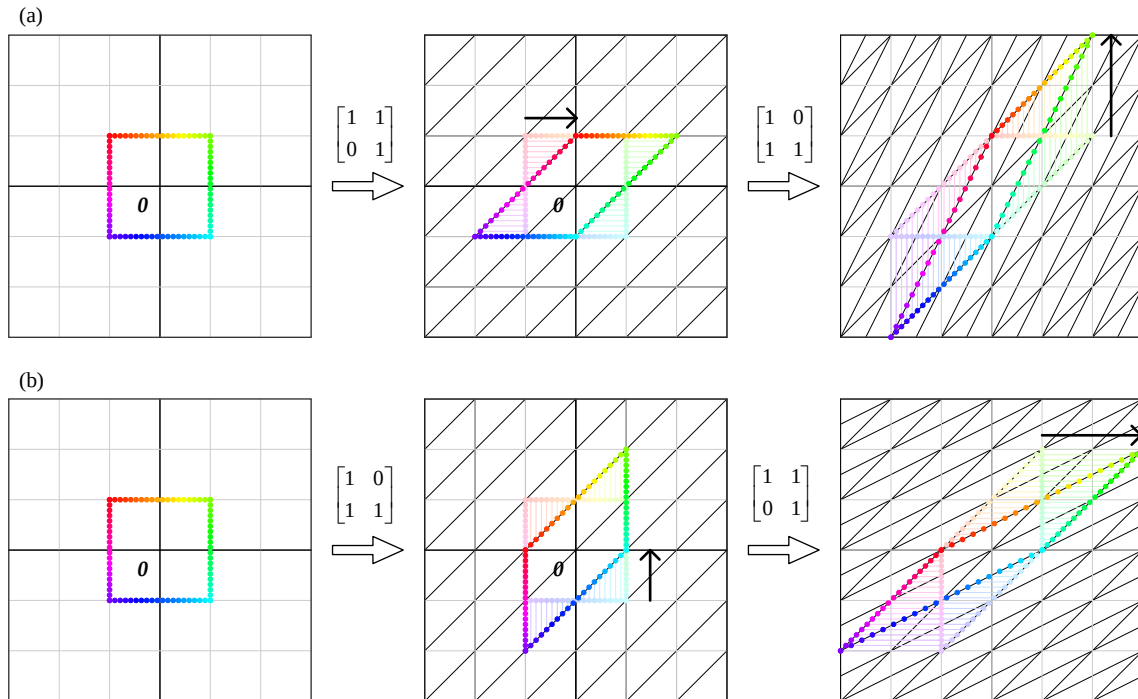


Figure 9. Consecutive Shears Along Different Axes in the Plane

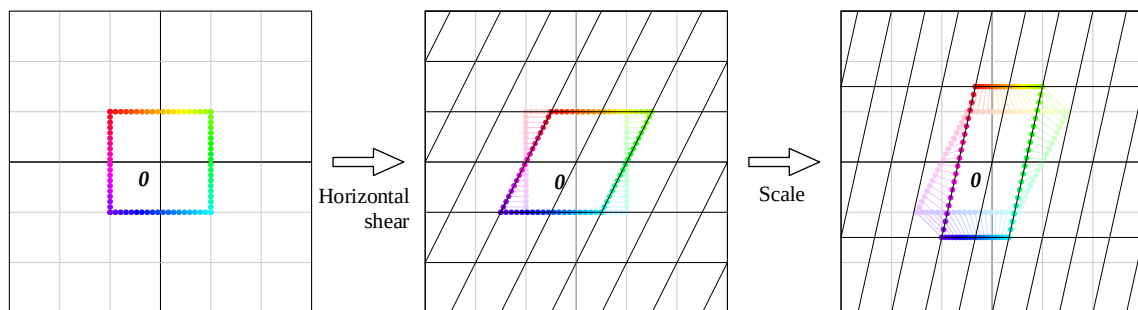
Scaling → Shear

For any 2×2 upper triangular matrix whose diagonal elements are all nonzero, the corresponding geometric operation can be written in the form "first horizontal shear (K) → then scaling (S)".

As an example, an upper triangular matrix can be written as " S (scaling) @ horizontal shear (K)":

$$\begin{bmatrix} 2/3 & 1/3 \\ 0 & 3/2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix}}_S \underbrace{\begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}}_K$$

As shown in Figure 10, SKx means that, for the column vector x , first K (horizontal shear) is applied, and then S (scaling).

Figure 10. First Horizontal Shear (K) → Then Scaling (S)

Similarly, for any 2×2 lower triangular matrix whose diagonal elements are all nonzero, the corresponding geometric operation can be written in the form "first vertical shear (K) → then scaling (S)".

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As an example, a lower triangular matrix can be written as " $\mathcal{S}$ (scaling) @ vertical shear ($\mathcal{K}$)":

$$\begin{bmatrix} 2/3 & 0 \\ 1/3 & 3/2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix}}_{\mathcal{S}} \underbrace{\begin{bmatrix} 1 & 0 \\ 2/9 & 1 \end{bmatrix}}_{\mathcal{K}}$$

As shown in Figure 11, $\mathcal{SK}\mathbf{x}$ means that, for the column vector $\mathbf{x}$, first $\mathcal{K}$ (vertical shear) is applied, and then $\mathcal{S}$ (scaling).

? This part will help us understand Cholesky decomposition and LDL decomposition.

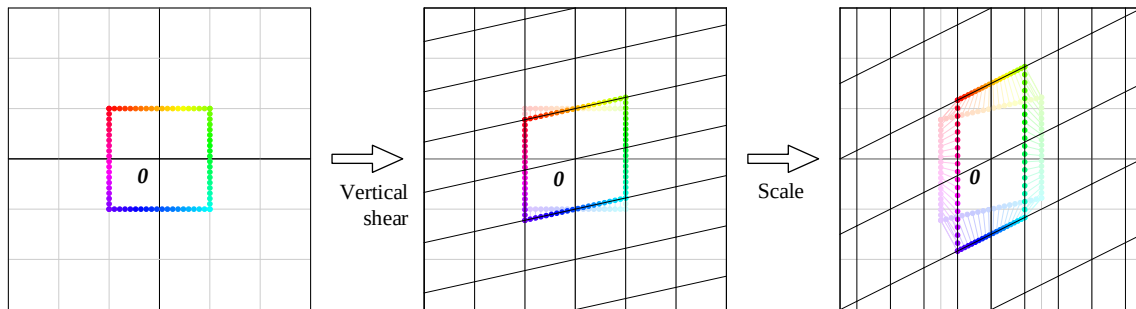


Figure 11. First Vertical Shear ($\mathcal{K}$) → Then Scaling ($\mathcal{S}$)

Shear → Scaling

Conversely, for any 2×2 upper triangular matrix whose diagonal elements are all nonzero, the corresponding geometric operation can also be written in the form "first scaling ($\mathcal{S}$) → then horizontal shear ($\mathcal{K}$)".

Using this example, the same upper triangular matrix can also be written as "horizontal shear ($\mathcal{K}$) @ $\mathcal{S}$ (scaling)":

$$\begin{bmatrix} 2/3 & 1/3 \\ 0 & 3/2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 2/9 \\ 0 & 1 \end{bmatrix}}_{\mathcal{K}} \underbrace{\begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix}}_{\mathcal{S}}$$

As shown in Figure 12, $\mathcal{KS}\mathbf{x}$ means that, for the column vector $\mathbf{x}$, first $\mathcal{S}$ (scaling) is applied, and then $\mathcal{K}$ (shear along the horizontal axis).

Comparing Figure 10 and Figure 12, we find the final results are the same.

⚠ Note that and are transposes of each other.

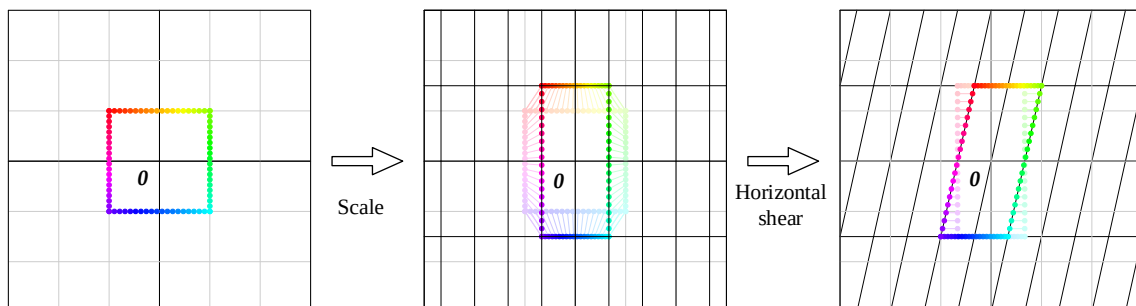


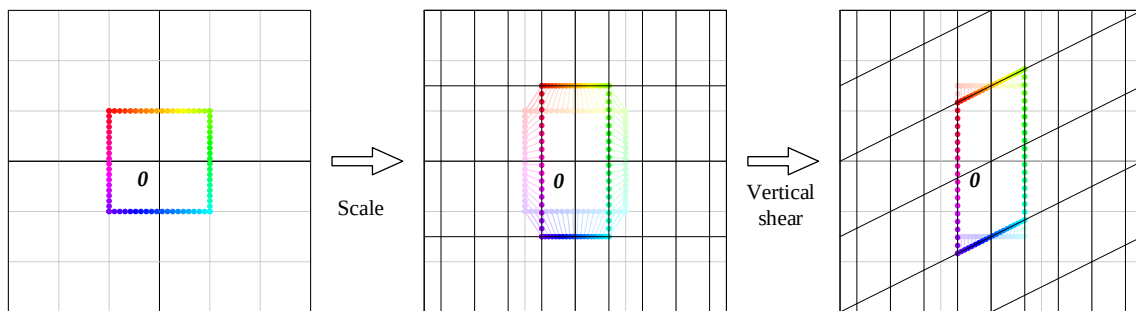
Figure 12. First Scaling ($\mathcal{S}$) → Then Horizontal Shear ($\mathcal{K}$)

Similarly, for any 2×2 lower triangular matrix whose diagonal elements are all nonzero, the corresponding geometric operation can be written in the form "first scaling ($\mathcal{S}$) → then vertical shear ($\mathcal{K}$)". Take the following example,

$$\begin{bmatrix} 2/3 & 0 \\ 1/3 & 3/2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}}_{\mathcal{K}} \underbrace{\begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix}}_{\mathcal{S}}$$

⚠ Note that and are transposes of each other.

Figure 13 And Figure 11 give the same result.

Figure 13. First Scaling ($\mathcal{S}$) → Then Vertical Shear ($\mathcal{K}$)

Shear Along the x_1 Axis in 3D Space

The shear transformations in 3D space can be viewed separately along three directions (the x_1 , x_2 , and x_3 axes); the shear along each direction can be further divided into the following two types. Let's look at them one by one.

Shear along the x_1 axis means that the value of x_1 changes as x_2 and x_3 change.

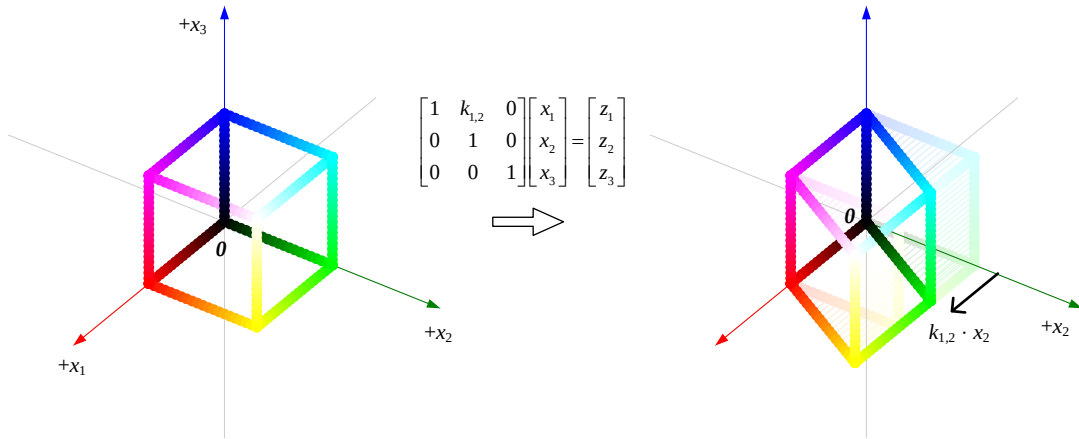
As shown in Figure 14, let's first look at how the shear value of x_1 changes with x_2 : the corresponding matrix multiplication is

$$\begin{bmatrix} 1 & k_{1,2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

It is easy to see that the determinant of the operation matrix is 1, meaning the volume does not change after the geometric transformation.

Expanding the equation above gives

$$\begin{cases} z_1 = x_1 + k_{1,2} \cdot x_2 \\ z_2 = x_2 \\ z_3 = x_3 \end{cases}$$

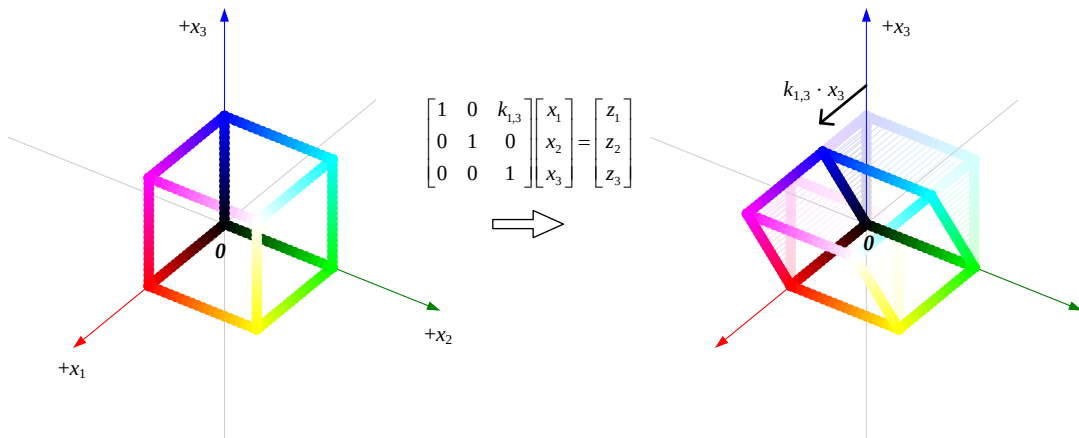
Figure 14. Shear Along x_1 , Relative to x_2 , in 3D Space

Now look at how the shear value of x_1 changes with x_3 : as shown in Figure 15, the corresponding matrix multiplication is

$$\begin{bmatrix} 1 & 0 & k_{1,3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding gives

$$\begin{cases} z_1 = x_1 + k_{1,3} \cdot x_3 \\ z_2 = x_2 \\ z_3 = x_3 \end{cases}$$

Figure 15. Shear Along x_1 , Relative to x_3 , in 3D Space

We find that these two matrix multiplications are commutative, that is,

$$\begin{bmatrix} 1 & k_{1,2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 & k_{1,3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & k_{1,3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & k_{1,2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & k_{1,2} & k_{1,3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This shows that the matrix multiplication of two same-axis shear operations is commutative; that is, in this special case, matrix multiplication satisfies the commutative property.

As shown in Figure 16, the corresponding matrix multiplication is

$$\begin{bmatrix} 1 & k_{1,2} & k_{1,3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding gives

$$\begin{cases} z_1 = x_1 + k_{1,2} \cdot x_2 + k_{1,3} \cdot x_3 \\ z_2 = x_2 \\ z_3 = x_3 \end{cases}$$

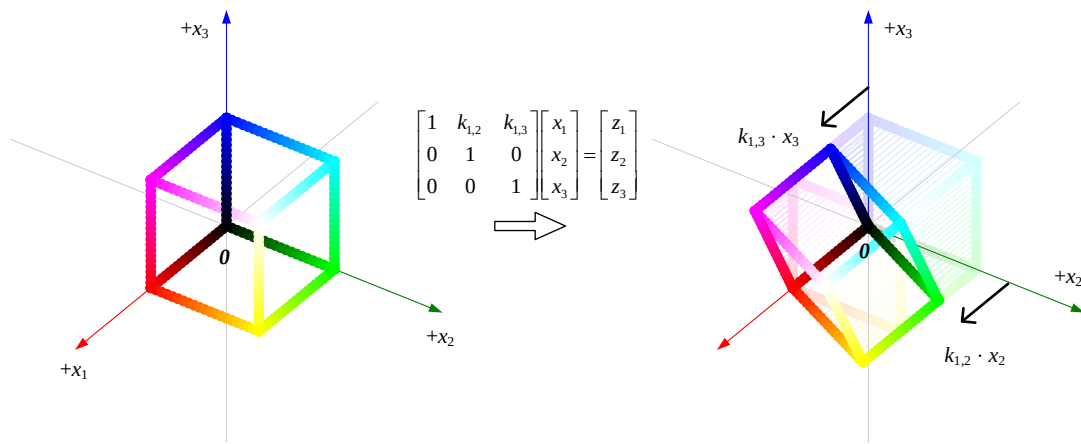


Figure 16. Shear Along x_1 , Relative to x_2 and x_3 , in 3D Space

Shear Along the x_2 Axis in 3D Space

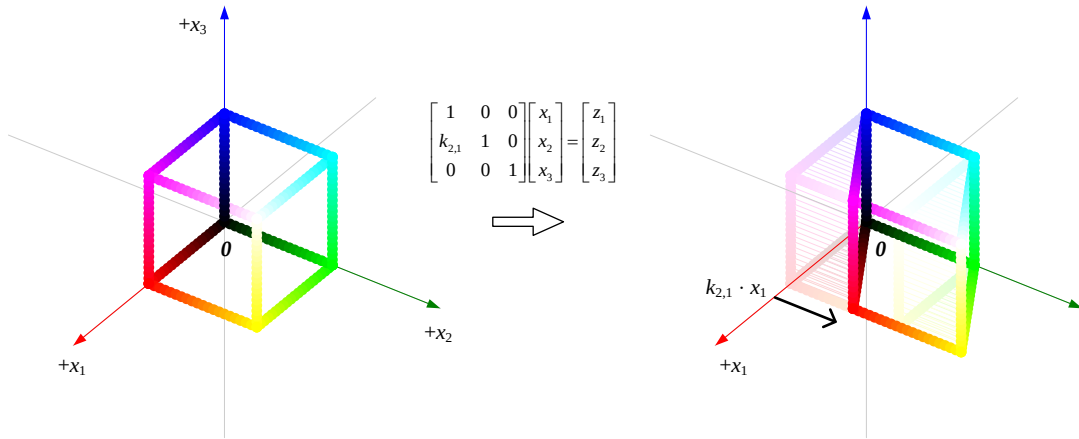
Similarly, the shear along the x_2 axis also has two references, x_1 and x_3 , shown respectively in Figure 17 and Figure 18.

As shown in Figure 17, let's first look at how the shear value of x_2 changes with x_1 : the corresponding matrix multiplication is

$$\begin{bmatrix} 1 & 0 & 0 \\ k_{2,1} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding the equation above gives

$$\begin{cases} z_1 = x_1 \\ z_2 = k_{2,1}x_1 + x_2 \\ z_3 = x_3 \end{cases}$$

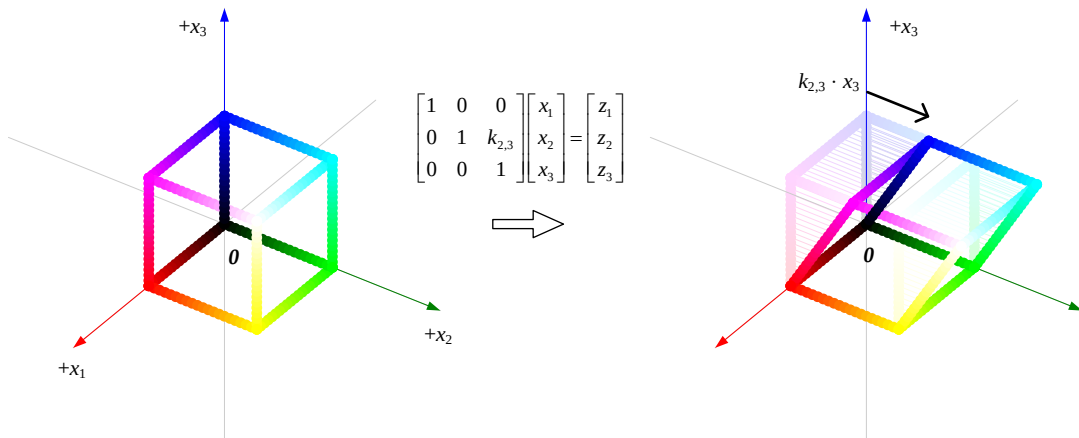
Figure 17. Shear Along x_2 , Relative to x_1 , in 3D Space

As shown in Figure 18, let's first look at how the shear value of x_2 changes with x_3 : the corresponding matrix multiplication is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & k_{2,3} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding the equation above gives

$$\begin{cases} z_1 = x_1 \\ z_2 = x_2 + k_{2,3}x_3 \\ z_3 = x_3 \end{cases}$$

Figure 18. Shear Along x_2 , Relative to x_3 , in 3D Space

Combining Figure 17 and Figure 18, we obtain Figure 19. The corresponding matrix multiplication is

$$\begin{bmatrix} 1 & 0 & 0 \\ k_{2,1} & 1 & k_{2,3} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

Expanding gives

$$\begin{cases} z_1 = x_1 \\ z_2 = k_{2,1} \cdot x_1 + x_2 + k_{2,3} \cdot x_3 \\ z_3 = x_3 \end{cases}$$

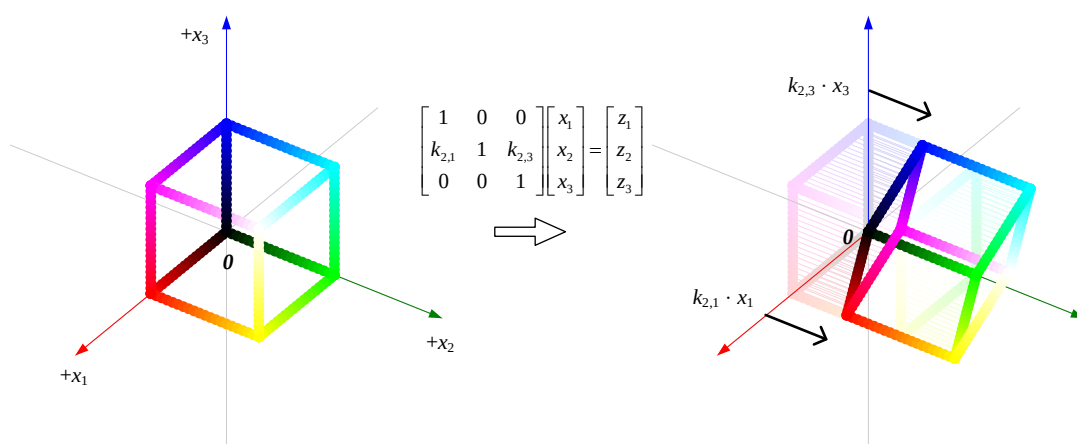


Figure 19. Shear Along x_2 , Relative to x_1 and x_3 , in 3D Space

Shear Along the x_3 Axis in 3D Space

Shear along the x_3 axis likewise has two references, x_1 and x_2 , shown as Figure 20 and Figure 21.

Please analyze these two figures on your own.

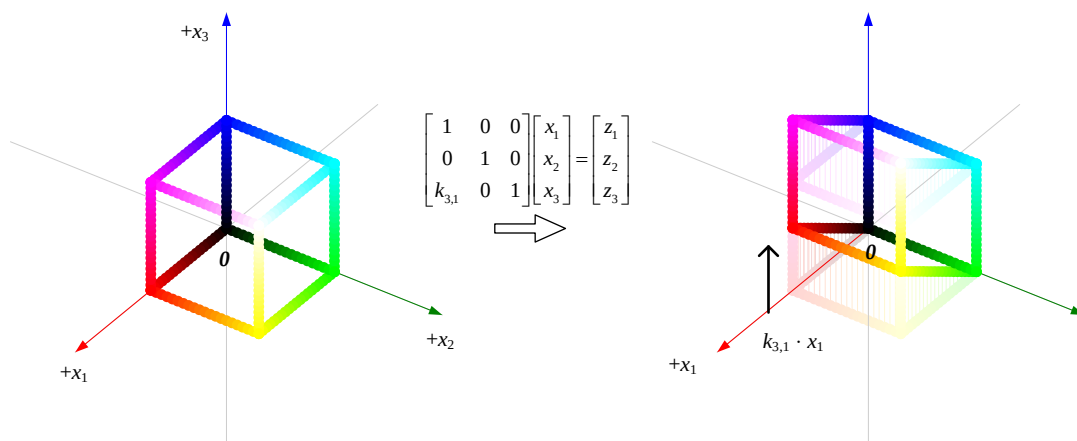
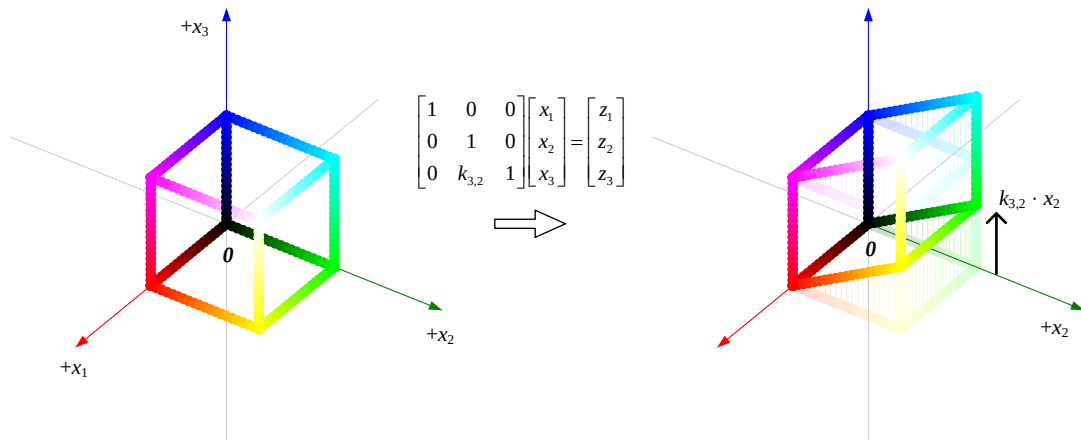


Figure 20. Shear Along x_3 , Relative to x_1 , in 3D Space

Figure 21. Shear Along x_3 , Relative to x_2 , in 3D Space

? Please multiply the two matrices corresponding to Figure 20 and Figure 21, and explain the corresponding geometric operation.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Which of the following matrices represent planar shears, and which do not?

▶ $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$

Q2. Given a set of planar point coordinates $\text{points} = [(1, 0), (0, 1), (-1, 0), (0, -1), (0, 0)]$, write Python

code to compute $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ the corresponding shear, and output the new coordinates after the shear.

Q3. Why does planar shear not change area?

Q4. Apply a horizontal shear to the point $(3, 4)$ with shear factor $k = 2$. Use matrix multiplication to compute the transformed coordinates.

Q5. Please compute the following chain of matrix multiplications, and try to explain the result from a geometric perspective.

► $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

► $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$

► $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

► $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} @ \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Q6. Write the following diagonal square matrix in both forms: "scaling @ shear" and "shear @ scaling".

► $\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$

► $\begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

Q7. Please modify LA_08_02_01.ipynb to visualize the planar shear operations from Figure 1 ~ Figure 9.

Q8. Please modify LA_08_03_01.ipynb to visualize the 3D shear operations from Figure 14 ~ Figure 21.

Q9. Please try to explain the geometric operation of the following square matrices in multidimensional space:

► $\begin{bmatrix} 1 & 0 & k & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

► $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & k & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

► $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ k & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$